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Regional adaptive hydropower-pumped storage collaborative planning for low-carbon transition: Flexibility enhancement and benefit evaluation based on resource endowment differences

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HIGHLIGHTS

- An innovative "three-step collaborative" modeling framework is proposed, linking long-term planning (MESSAGEix), hourly operational simulation (DISPASET), and comprehensive evaluation (CRITIC-TOPSIS).
- Quantifies the superior benefits of variable-speed pumped storage in enhancing flexibility, reducing loss-of-load hours by up to 166 h and increasing renewable energy utilization by 17.8%.
- Systematically compares planning strategies for hydropower flexibility retrofitting and pumped storage at the provincial level for regions with different water resource endowments for the first time.
- Determines the required scale of pumped storage (e.g., 63–72 GW) to ensure grid reliability for water-scarce regions under deep decarbonization scenarios.
- Provides data-driven, regionally-tailored planning strategies to support a stable and economical low-carbon transition of power systems.

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ABSTRACT

Large-scale integration of wind and solar power combined with the progressive decommissioning of conventional coal-fired power plants poses significant challenges to power system flexibility. Although hydropower and pumped storage can provide flexible regulation, China's regional water resource endowments vary significantly, and their impacts on low-carbon transition across different regions have not been systematically quantified. Therefore, this study proposes a three-step collaborative modeling framework, developing a MESSAGEix power generation system planning model aimed at minimizing system costs and a DISPASET unit commitment optimization model that precisely captures hydropower unit operational characteristics, proposing differentiated hydropower flexibility retrofitting and pumped storage collaborative planning schemes. A multi-dimensional comprehensive evaluation index system encompassing economic, technical, environmental, and social dimensions was established, with planning schemes for various provinces scientifically evaluated through the CRITIC-TOPSIS multi-criteria decision-making method. Results show hydropower flexibility retrofitting reduces loss-of-load hours in Xinjiang and Shaanxi by 6 h and 20 h respectively, while variable-speed pumped storage achieves greater reductions of 40 h and 166 h, both improving renewable utilization by 0.6%–17.8%. By 2030, variable-speed schemes perform best for Xinjiang and Shaanxi, while Yunnan's basic scheme excels; by 2060, incremental pumped storage schemes for Xinjiang and Shaanxi, and variable-speed scheme for Yunnan demonstrate optimal performance.This research provides theoretical support and empirical evidence for differentiated hydropower and pumped storage planning

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1. Introduction

1.1. Background

The global transition toward low-carbon power systems is characterized by two concurrent phenomena: the accelerated decommissioning of conventional coal-fired power plants and the substantial increase in variable renewable energy generation, particularly from wind and solar sources. This transition, while environmentally beneficial, potentially creates a significant deficit in system flexibility regulation resources, thereby compromising grid stability and operational reliability [1]. To mitigate these challenges, enhancing power grid flexibility regulation capabilities through strategic optimization of peak-shaving resources and energy storage technologies has emerged as a critical imperative for maintaining system stability [2,3]. Within this context, hydropower and pumped storage hydropower (PSH) technologies are preferentially deployed as primary large-scale flexibility regulation resources due to their technological maturity, scalable capacity, and favorable economic characteristics when compared to emerging alternatives such as hydrogen storage and compressed air energy storage systems. Nevertheless, China exhibits pronounced heterogeneity in regional water resource endowments, resulting in substantial variations in hydropower development potential and pumped storage construction conditions across different geographical areas [4]. Consequently, a comprehensive understanding of regional resource endowment differences is essential for optimizing the flexibility enhancement potential of various hydropower and pumped storage transition schemes. Furthermore, elucidating the complex coordination mechanisms between hydropower, pumped storage, conventional thermal generation, and renewable energy sources under diverse boundary constraints during sequential stages of power system transition is fundamental to achieving a stable, reliable, and sustainable low-carbon transformation of China's power infrastructure.

1.2. Problem statement

Hydropower flexibility retrofitting and variable-speed pumped storage hydropower technology are two core means to enhance system flexibility, as shown in Fig. 1. For existing hydropower stations, since the installed capacity is approaching the development limit, flexibility enhancement mainly relies on technical retrofitting [5], such as upgrading turbine runners and governing systems [6], and optimizing control processes [7,8]. These measures aim to expand the operating range of hydropower units and improve their response speed. For pumped storage units, the focus has shifted from vigorously developing fixed-speed units to emphasizing the application of variable-speed units. Compared to fixed-speed pumped storage, variable-speed pumped storage can adapt to larger speed fluctuations [9], improving the generation and pumping efficiency of pumped storage units. Although hydropower flexibility retrofitting (such as extending the lower output limit of hydropower units to 0%) and variable-speed pumped storage are core measures to enhance system flexibility, their actual flexibility enhancement effects are limited by regional water resource endowments and power structure differences. To comprehensively quantify and evaluate the potential and limitations of these measures, the following questions need to be further addressed:

1. Under regional resource endowment constraints, how should hydropower flexibility retrofitting and variable-speed pumped storage be differentially configured to match coal power retirement sequences and renewable energy grid integration scales?
2. How can the comprehensive impacts of these two measures on system flexibility, economics, and social and environmental aspects in 2030 (transition period) and 2060 (deep decarbonization period) be comprehensively evaluated?

1.3. Literature review

To clearly define the knowledge gap, this study critically reviews the literature from three key perspectives: hydropower flexibility enhancement, long-term transition planning, and comprehensive evaluation methodologies.

Existing research extensively demonstrates that hydropower flexibility is crucial for integrating variable renewables. Studies have provided valuable frameworks for assessing hydropower flexibility at the basin level [10] and optimizing the joint operation of multi-energy systems [11–13]. Methodologies have also been developed to manage short-term risks and enhance peak-shaving performance in hydro-wind-solar hybrid systems [14,15]. However, a significant limitation pervades these studies: they predominantly focus on plant- or basin-level analyses, where resource endowments are relatively homogeneous. There is a notable lack of systematic comparative analysis at the provincial scale, which is essential for national-level policymaking, especially concerning how flexibility retrofitting strategies should differ in regions with vastly

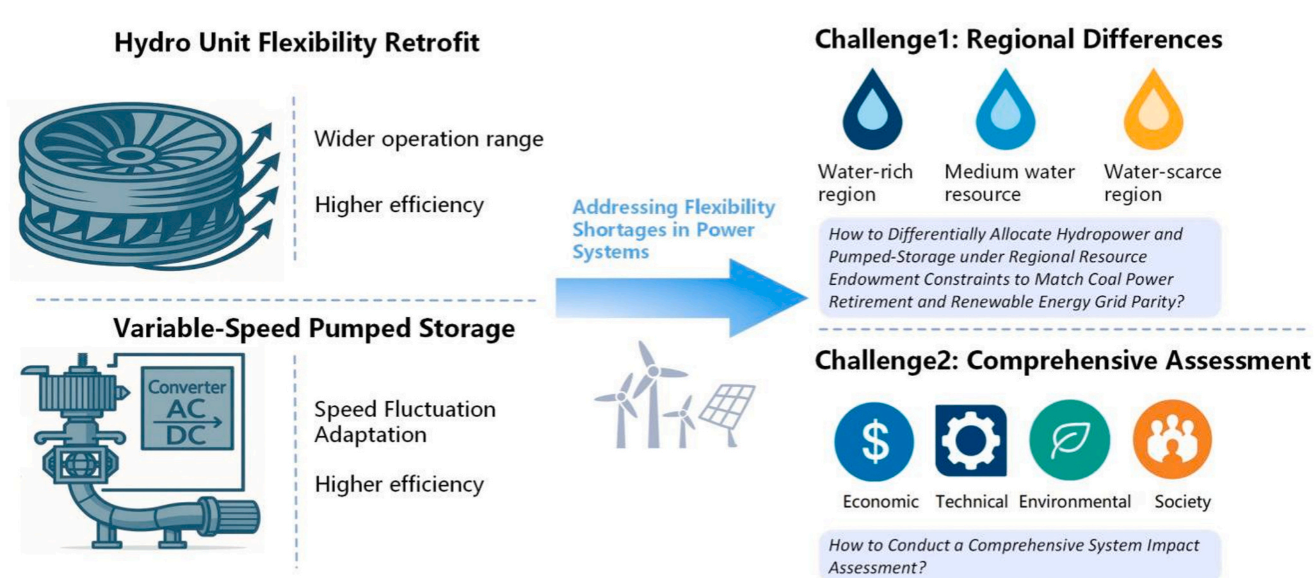

Fig. 1. Hydropower and pumped storage flexibility enhancement measures and research challenges.

different water and renewable resource endowments. This leaves a critical gap in understanding how to tailor flexibility solutions to specific regional contexts.

From a long-term planning perspective, the indispensable role of hydropower and pumped storage in facilitating the energy transition is well-established. Researchers have investigated the system-level effects of flexible hydropower operation under various coal power retirement scenarios [16], explored the market competitiveness impacts of carbon trading mechanisms [17], and quantified the low-carbon attributes of pumped storage through life-cycle analysis [18,19]. Nevertheless, these transition studies often employ simplified or aggregated models for hydropower and pumped storage technologies. They typically lack the granular representation needed to differentiate between various technology options, such as fixed-speed versus variable-speed pumped storage, or the specific operational characteristics of retrofitted hydropower units. This methodological simplification prevents a precise quantification of the distinct flexibility benefits each specific technology can provide, thereby obscuring their true value in deep decarbonization pathways.

Establishing a scientific evaluation framework is fundamental to assessing energy system planning schemes. A variety of multi-criteria decision-making (MCDM) frameworks have been proposed, considering dimensions such as economy, environment, energy efficiency, and reliability [20–29]. These studies have utilized methods like objective allocation, cloud models, and various weighting techniques to handle the uncertainties and complexities inherent in system evaluation. Despite these advances, a unified and comprehensive evaluation framework specifically designed for assessing hydropower's role in the low-carbon transition is still missing. Existing frameworks often neglect to integrate key technical flexibility indicators with broader economic, social, and environmental impacts in a single, cohesive system. This absence of a holistic evaluation tool makes it difficult to comprehensively measure and compare the multi-faceted benefits of different hydropower and pumped storage planning schemes.

1.4. Contributions

To address the gaps in existing research, this paper conducts a multi-provincial comparative analysis focusing on scenario setting and systematic quantitative assessment of hydropower flexibility operation enhancement in regions with different water resource endowments. Three representative provinces (Xinjiang, Shaanxi, and Yunnan) are strategically selected to represent China's primary low-carbon transition pathways based on their distinct resource endowment characteristics. It establishes a power generation system planning model and a power system unit combination model that includes hydropower unit operation characteristics, and constructs a complete comprehensive evaluation system for power generation system low-carbon transition. The main contributions of this paper are as follows:

1. This study develops an application-oriented collaborative modeling framework that soft-links the MESSAGEix planning model with the DISPASET unit commitment model, enabling multi-scale analysis from macro planning to micro operation. Nine hydropower and pumped storage planning schemes have been designed to address the limitations of existing research that typically focuses only on single plant or basin-level analysis.

2. Hydropower unit operational characteristics have been parameterized within the DISPASET model through specific parameters (0% minimum output limit and 45% rated capacity/min ramping rate). This approach enables the first quantitative analysis of differences between fixed-speed and variable-speed pumped storage technologies, providing methodological support for accurate flexibility assessment.

3. A comprehensive evaluation system based on the CRITIC-TOPSIS method has been developed that integrates economic, technical, environmental, and social indicators into a unified framework. This objective weighting method quantifies the emphasis of each scheme on different indicators, providing scientific decision-making basis for regionally differentiated hydropower and pumped storage planning.

The structure of this paper is as follows: Section 2 establishes the power generation system planning model and the power system unit combination model, and introduces the specific content of the comprehensive evaluation index system. Section 3 takes different schemes of hydropower and pumped storage planning as the research object, exploring the flexibility indicators and comprehensive evaluation indicators of hydropower and pumped storage under different scenarios. Sections 4 and 5 discuss and summarize the content of the entire paper.

2. Method

2.1. Collaborative modeling framework

This study introduces a three-step collaborative modeling framework, depicted in Fig. 2. The process begins with the MESSAGEix model, which optimizes the regional energy installation structure for target years 2030 (transition phase) and 2060 (deep decarbonization). Its main goal is to minimize the overall discounted system cost across the planning horizon, considering constraints such as water availability, clean energy installation ratios, annual carbon emission limits, and pumped storage requirements. The results provide essential data on installed capacities for further analysis. Next, the DISPASET model builds on the planning outcomes from MESSAGEix, focusing on the flexibility of hydropower units to lower operational costs. Operating on a 1-h optimization interval, DISPASET captures operational details like output efficiency, minimum startup times, operational ranges, startup costs, and reservoir characteristics. It simulates a full year of power system operations (8760 h) to evaluate performance and flexibility across varied scenarios. Finally, a comprehensive evaluation index system is developed, based on four dimensions: economy (construction and operating costs), environment (CO2 emissions and renewable energy use), technology (loss-of-load probabilities and load variability), and social impacts (employment and regional development). The CRITIC-TOPSIS multi-criteria decision-making approach is utilized to objectively assess and rank different schemes, providing a well-rounded decision-making framework, see Appendix A. This integration of three phases ensures a cohesive analytical process, where each phase's output serves as critical input for subsequent analyses.

While MESSAGEix and DISPA-SET are both mature open-source platforms, their soft-linking for hydropower and pumped storage planning involves three recognized technical bottlenecks that go beyond routine data interfacing. First, cross-scale temporal consistency requires bridging the 15-year planning horizon of MESSAGEix with the 8760-h operational resolution of DISPA-SET; in our implementation, approximately 15% of initial planning solutions were found operationally infeasible and required iterative feedback adjustment between the two models to achieve convergent results. Second, variable-speed pumped storage technology is not natively supported in DISPA-SET, necessitating non-trivial modifications to the core unit commitment constraints, including bidirectional power regulation range (±1 operational range), enhanced ramping capability (67%/min versus 25%/min for fixed-speed units), and technology-specific reservoir balance formulations. Third, hydropower flexibility retrofit parameterization required calibration against actual engineering data from completed retrofit projects (e.g., Xiluodu and Xiangjiaba stations), where ramping rates improved from 1 to 2%/min to 45%/min and minimum output limits were extended from 40% to 0% of rated capacity. These implementation challenges, while characteristic of application-oriented system integration, represent concrete technical obstacles that have limited quantitative flexibility assessment of advanced pumped storage technologies in prior regional planning studies.

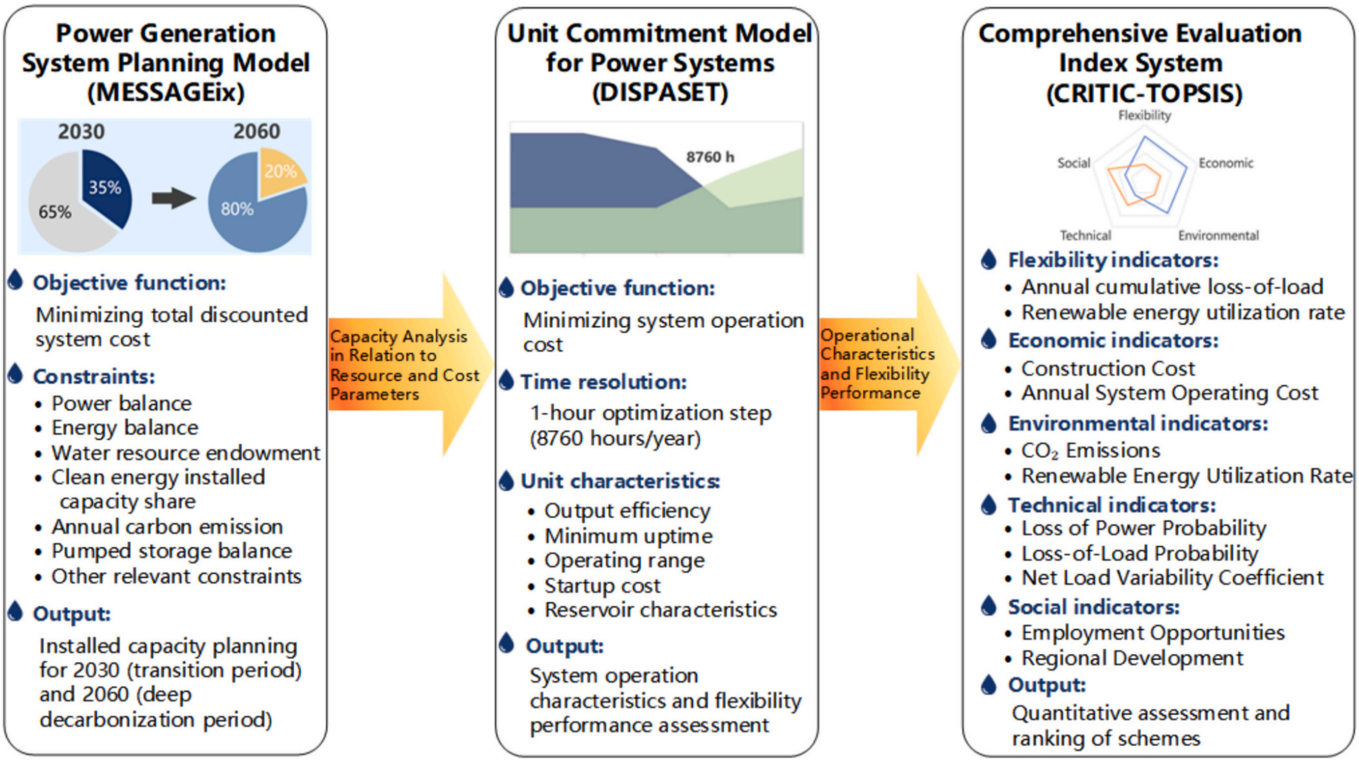

Fig. 2. Methodological technical roadmap.

2.2. Power generation system planning model

The power generation system planning model optimizes energy system planning through a mixed-integer linear optimization framework, integrating technical, economic, and environmental factors. This model determines the optimal energy supply structure and technology transition pathways to meet energy demand at minimum cost [30].

2.2.1. Objective function

The objective function minimizes total system costs, including generation costs, fixed operational costs, and variable operation and maintenance costs, using discounting methods for residual values [31]:

$$\min C_1 = \sum_{y=1}^{N} \left(C_{1y} + C_{2y} + C_{3y}\right)(1+r)^{1-y} \tag{1}$$

where represents the system cost minimization objective. Detailed cost component definitions (eqs. 2–5) and parameter specifications are provided in Supplementary Material A.

2.2.2. Key constraints

Standard power and energy balance constraints ensure supply-demand equilibrium (detailed formulations in Supplementary Material A). The key constraints specific to our hydropower and pumped storage planning framework include:

Water resource endowment constraint.

Hydropower resource constraints reflect regional water availability and sustainability limits. This study considers varying exploitable potentials, mandating that total installed capacity aligns with resource potential and technical conditions, while annual additions stay within maximum construction limits, as specified below.

$$\sum_{w\in S} X_{w,y} \le X_{w,y}^{total} \tag{2}$$

$$\underset{w\in S}{X_{w,y}} \le \underset{w\in S}{X_{w,y}^{\lim}} \tag{3}$$

$$\underset{w\in S}{X_{w,y}} \le \left(1 + \underset{w\in S}{\lambda_{w,y}}\right) X_{w,y-1} \tag{4}$$

where S denotes the set of hydropower technologies across all years of the planning period; $X_{w,y}$ represents the newly installed capacity of hydropower technology in year y;$X_{w,y}^{total}$ indicates the technically exploitable potential of water resources in the region; $X_{w,y}^{\lim}$ specifies the annual capacity addition limit in year y;$\lambda_{w,y}$ denotes the capacity growth rate in year y;$X_{w,y-1}$ refers to the total installed capacity in the year preceding the planning period.

2.3. Clean energy installed capacity share constraint

The clean energy capacity share constraint ensures future expansion meets clean energy targets, driving energy mix transformation, reducing traditional energy reliance, and supporting sustainability and emission goals, as shown in eq. 5.

$$\sum_{r\in KA} K_{r,y} \ge \partial_y \sum_{r\in A} K_{r,y} \tag{5}$$

where KA denotes the set of clean energy sources (e.g., wind and solar), and ∂_y represents the clean energy generation share coefficient in year y of the planning period.

2.4. Annual carbon emission constraint

To facilitate low-carbon transition of power systems, the annual carbon emission constraint serves as a fundamental modeling requirement, ensuring progressively reduced emission levels each year, as shown in eq. 6.

$$\sum_{f\in HA} K_{f,y}\eta_{f,y} \le e_{\max}^{y} \tag{6}$$

where HA represents the set of fossil fuel power generation technologies,$\eta_{f,y}$ denotes the carbon emission factor of generation technology, and $e_{\max}^{y}$ specifies the maximum allowable carbon emissions.

2.5. Pumped storage balance constraint

Pumped hydro balance constraints govern the equilibrium control of energy storage and discharge processes. Proper scheduling, technical/safety controls, and environmental measures ensure stable system operation for reliable power and storage services. Development is further limited by regional resource availability, as expressed in the following equation.

$$\sum_{h\in psh} X_{h,y} \le X_{h,y}^{total} \tag{7}$$

$$\underset{h\in psh}{X_{h,y}} \le \underset{h\in psh}{X_{h,y}^{\lim}} \tag{8}$$

$$K_{h,y}^{storage} = \sum_{h\in psh} K_{h,y}^{tur} - \sum_{h\in psh} K_{h,y}^{cha} \tag{9}$$

$$\sum_{h\in psh} K_{h,y} = \sum_{h\in psh} K_{h,y-1} + K_{h,y}^{storage} \tag{10}$$

where psh denotes the set of pumped storage power generation technologies;$X_{h,y}$ represents the installed pumped storage capacity in year y of the planning period;$X_{h,y}^{total}$ indicates the regional upper limit for pumped storage capacity;$X_{h,y}^{\lim}$ stands for the annual capacity constraint of pumped storage in planning year y;$K_{h,y}^{storage}$ signifies the stored energy of pumped storage during year y;$K_{h,y}^{tur}$ and $K_{h,y}^{cha}$ denotes the pumping volume and power generation of pumped storage in year y of the planning period respectively;$K_{h,y}$ represents the final stored energy of pumped storage at the end of year y in the planning period;$K_{h,y-1}$ refers to the stored energy of pumped storage in the year preceding the planning period.

Additional standard constraints including capacity growth rate limits and installation bounds are detailed in Supplementary Material A.

2.6. Unit commitment model for power systems

To analyze the operational performance of different hydropower and pumped storage planning schemes under the context of enhancing power system flexibility, this paper establishes a unit commitment model for power systems. The model aims to minimize system operating costs, employs a 1-h optimization time step, and meticulously captures the operational characteristics of each generating unit, including output efficiency, minimum uptime, operating range, startup costs, and reservoir characteristics.

2.6.1. Objective function

The objective function of the model is formulated to minimize power system operating costs, which include startup costs, shutdown costs, fixed costs, variable costs, ramping costs, and loss-of-load costs [32].

$$C_{min} = C_{i,u,s} + C_{i,u,d} + C_{i,u,f} + C_{i,u,v} + C_{i,u,r} + C_{i,u,l} + C_{i,u,o} \tag{11}$$

where C_{min} represents the system cost minimization objective; subscript i denotes the i-th time step sequence; subscript u indicates the set of different generator types; $C_{i,u,s}$,$C_{i,u,d}$,$C_{i,u,f}$,$C_{i,u,v}$,$C_{i,u,r}$,$C_{i,u,l}$ and $C_{i,u,o}$ represent respectively the startup cost, shutdown cost, fixed cost, variable cost, ramping cost, downward regulation cost, and loss-of-load cost incurred by unit u during time step i. All cost parameters are expressed in units of 10^4 CNY/MW, with specific values provided in Table 2.

2.6.2. Constraint condition

2.6.2.1. Power system supply-demand balance constraint. The power system supply-demand balance constraint constitutes a fundamental principle of power system operation. It requires that the total load demand minus the total loss-of-load in each region must equal the sum of active power generated by all types of generation units within that region and the power transmitted from other regions, as shown in eq. 20 [33].

$$P_i^D - P_i^L = P_{i,u}{}^p + T_i \tag{12}$$

where P_i^D represents the total load demand in the region at time i; P_i^L denotes the system loss-of-load at time i; $P_{i,u}{}^p$ signifies the total active power output from all generation units in the region at time i; T_i indicates the net power transmission at time i. This study accounts for inter-provincial power exchange through a simplified but realistic approach that reflects China's current transmission mechanism. Historical net inter-provincial transmission volumes are extracted from China Electric Power Yearbook (annual totals) and allocated to 8760 hourly time series using typical load curves. Net imports reduce provincial load, while net exports increase it. This approach reasonably reflects China's current inter-provincial transmission mechanism, which operates primarily through medium-to-long-term contract trading with predetermined annual delivery plans, rather than real-time market bidding.

2.7. Generator power output constraints

Generator power output constraints refer to the operational limitations imposed on the power output range of generating units in power systems. As shown in eq. 21 [34], the power output of generating units must exceed the minimum stable output level specified for each type of generator while remaining below the maximum output limit (typically the rated generation capacity of the unit).

$$P_{i,u}^{\min} < P_{i,u}^{P} < P_{i,u}^{\max} \tag{13}$$

where $P_{i,u}^{\min}$ and $P_{i,u}^{\max}$ represent the minimum and maximum output power of different types of generating units at time i, respectively.

Table 1
Different flexibility enhancement ccenarios in three provinces.

Region	Year	Scheme	Scheme Description
Xinjiang	2030	XJ-A	20GW coal power retirement
		XJ-B	Hydropower ramping: 45%, operating lower limit: 0
		XJ-C	Convert fixed-speed PSH units to variable-speed units
Shaanxi	2030	SX-A	15GW coal power retirement
		SX-B	Hydropower ramping: 45%, operating lower limit: 0
		SX-C	Convert fixed-speed PSH units to variable-speed units
Yunnan	2030	YN-A	Unchanged
		YN-B	Hydropower ramping: 45%, operating lower limit: 0
		YN-C	Convert fixed-speed PSH units to variable-speed units
Xinjiang	2060	XJ-A-2	Unchanged
		XJ-B-2	Hydropower ramping: 45%, operating lower limit: 0
		XJ-C-2	Convert fixed-speed PSH units to variable-speed units
Shaanxi	2060	SX-A-2	Unchanged
		SX-B-2	Hydropower ramping: 45%, operating lower limit: 0
		SX-C-2	Convert fixed-speed PSH units to variable-speed units
Yunnan	2060	YN-A-2	Unchanged
		YN-B-2	Hydropower ramping: 45%, operating lower limit: 0
		YN-C-2	Convert fixed-speed PSH units to variable-speed units

Table 2
System cost parameters for different power generation technologies (unit: 10^4 CNY/MW).

Power Generation Type and Region	Start-up Cost	Maintenance Cost	Variable Cost	Peak-shaving Cost	Load of Loss Cost
Hydropower	0	25.94	0	0	–
Photovoltaic	0	6.75	0	0	–
Wind Power	0	14.3	0	0	–
Coal Power	0.11	6.1	0.0264	0.0031	–
Pumped Storage	0	14.82	0	0	–
Xinjiang	–	–	–	–	0.4452
Shaanxi	–	–	–	–	0.1741
Yunnan	–	–	–	–	1.2110

2.8. Generator ramping constraints

Generator ramping constraints refer to the rate-of-change limitations imposed on the power output of generating units during startup or shutdown processes in power systems. These constraints ensure that generating units adjust their power output in a safe and controlled manner during startup or shutdown, preventing instability or damage to the power system. This is illustrated in eq. 22.

$$R_u^{dowm} \le R_{u,i} - R_{u,i-1} \le R_u^{up} \tag{14}$$

where R_u^{dowm} and R_u^{up} represent the downward and upward ramping rate limits for generation technology units respectively;$R_{u,i}$ and $R_{u,i-1}$ denote the downward and upward ramping rates of unit u between time i and the previous time interval.

2.9. Carbon emission constraints

Power system carbon emission constraints establish limits on electricity-related carbon emissions to drive low-carbon development, reduce greenhouse gases, and achieve sustainability goals through regional emission caps, while restricting coal-fired generation output as shown in eq. 23.

$$F_l P_u^c \le F_n^{\max} \tag{15}$$

where F_l represents the unit emission coefficient of coal-fired generation; ${P_u}^c$ denotes the total power output of coal-fired units in the region; $F_n^{\max}$ indicates the carbon emission cap for the region.

Hydropower and pumped storage operational constraints

(1) Reservoir storage capacity constraints

Reservoir storage capacity constraints refer to the operational limitations imposed on water storage volumes during reservoir operation and management. These constraints ensure that water level adjustments and discharge scheduling across different time periods maintain reservoir operational safety while meeting water resource utilization and management requirements.

$$S_{min} \le S_i \le S_{max} \tag{16}$$

where S_{min} represents the minimum operational storage capacity for power generation;S_{max} denotes the maximum operational storage capacity for power generation; S_i indicates the reservoir storage capacity at time i.

(2) Reservoir power balance constraint

Reservoir power balance constraints regulate water levels and discharges to maintain electricity supply-demand equilibrium, ensuring safe operations while meeting dispatch requirements. The storage balance is formulated in energy-equivalent units (MWh), where all water-related terms are converted to their corresponding energy equivalents via the hydraulic head relationship. Storage capacity must satisfy:

$$S_i = S_{i-1} + Q_i^s \eta_C - S_i^s - S_i^x + S_i^r - P_i^s / \eta_G \tag{17}$$

S_{i-1} is the reservoir storage capacity at time *i-1; Q_i^s* is the actual pumping energy input of the unit at time *i; η_C* is the pumping efficiency of the unit; S_i^s is the energy equivalent of spilled water at time *i; S_i^x* is the reservoir discharge flow at time *i; S_i^r* is the energy equivalent of upstream runoff inflow at time *i; P_i^s* is the power generation of the unit at time *i; η_G* is the generation efficiency of the unit. Consequently, the term $Q_i^s \eta_C$ represents the effective energy stored from pumping (electrical input discounted by pump losses), while P_i^s/η_G represents the energy consumed from storage for generation (electrical output grossed up by generator losses).

(3) Reservoir storage constraints for pumped storage units in generation mode

Pumped storage units generate electricity by releasing water from upper reservoirs. Sustainable operation requires maintaining water volumes above minimum generation capacity levels. These reservoir storage constraints ensure continuous power production meets operational requirements.

$$\left(P_i^h / \eta_g\right) + S_i^x + S_i^s - S_i^r \le S_i \tag{18}$$

where P_i^h represents the power generation of the pumped storage unit at time i; η_g denotes the generation efficiency of the pumped storage unit.

(4) Pumping mode operational constraints

Pumped storage units operate in pumping mode by converting electricity to potential energy through water elevation. This energy-intensive process requires adequate reservoir capacity for water storage. Operational constraints ensure sufficient storage volume availability.

$$\eta_c Q_i - S_i^x - S_i^s + S_i^r \le S_i^c - S_i \tag{19}$$

where η_c represents the pumping efficiency of the pumped storage unit; S_i^c indicates the remaining storable energy capacity in the reservoir at time *i*.

2.10. Comprehensive evaluation index system

This study evaluates power system flexibility under different scenarios using stability and renewable integration metrics. System stability is measured by annual loss-of-load hours and cumulative loss-of-load, reflecting supply reliability. Renewable integration is assessed through wind and PV utilization rates, indicating system accommodation capacity. Specific metrics include:

Economic indicators

(1) Construction Cost (A11)

The initial investment cost for power generation technologies includes one-time capital expenditures during project construction, such as infrastructure and equipment costs.

$$A_{11} = \sum C_u \varepsilon_u \tag{20}$$

where C_u represents the newly added or retrofitted capacity for different generator types, and ε_u denotes the unit cost for capacity addition or retrofit.

(2) Annual System Operating Cost (A12)

This refers to the total expenses incurred by various power generation technologies within a year to ensure the normal operation of the power system. It includes startup costs, shutdown costs, fixed costs, variable costs, ramping costs, and loss-of-load costs.

$$A_{12} = C_{min} = C_{i,u,s} + C_{i,u,d} + C_{i,u,f} + C_{i,u,v} + C_{i,u,r} + C_{i,u,l} + C_{i,u,o} \quad (21)$$

2.11. Environmental indicators

The power sector, being China's primary CO_2 emission source, necessitates rigorous environmental evaluation for transition planning. This study employs two core metrics: CO_2 emissions and renewable utilization rate. These indicators enable informed decision-making for sustainable power system transformation.

(1) CO_2 Emissions (A21)

Refers to the total carbon dioxide emissions generated by the system during the calculation period. The equation is as follows:

$$A_{21} = \alpha_{21} W_l \quad (22)$$

here α_{21} denotes the CO_2 emission factor of coal power, set to 0.824 t/MWh in this study. This value is derived from the national CO_2 emission coefficient per unit of thermal power generation reported in the China Electric Power Industry Annual Development Report 2023 (824 g/kWh, equivalent to 0.824 t/MWh), which is based on nationwide operational statistics of thermal power units and is therefore widely adopted for power-sector carbon accounting due to its representativeness and timeliness. W_l denotes the total coal power capacity consumed by the system.

(2) Renewable Energy Utilization Rate (A22)

The ratio of practically utilized renewable electricity (total generation minus curtailed wind/solar power) to the total potential generation from renewable sources (such as wind and solar) in the power system. The equation is:

$$A_{22} = A_{ws} = \frac{P_u^a - P_u^q}{P_u^a} \quad (23)$$

2.12. Technical indicators

Technical indicators serve as critical metrics for evaluating system performance requirements in renewable-dominated power systems. This study selects three key indicators: Loss of Power Probability (LOPP), Loss-of-Load Probability (LOLP), and Net Load Variability Coefficient, as detailed below:

(1) Loss of Power Probability (A31)

This metric quantifies the time proportion during which the power system fails to meet load demand within a given period, calculated in this study using chronological simulation. The equation is:

$$A_{31} = \frac{\sum_{i=0}^{8760} N_{i,LPSP}}{N} \quad (24)$$
$$if\left(P_i^o > 0, N_{i,LPSP} = 1\right)$$

where $N_{i,LPSP}$ represents the number of power deficiency occurrences, and N denotes the total number of time intervals in the study period.

(2) Loss-of-Load Probability (A32)

The ratio of total generation deficit to total load demand. While power deficiency rate focuses on the duration of supply shortages, LOLP quantifies the magnitude of loss-of-load. Both serve as critical reliability metrics for power system supply adequacy. The equation is:

$$A_{32} = \frac{\sum_{i=0}^{8760} P_i^o}{P_i^D} \quad (25)$$
$$if\left(P_i^o > 0\right)$$

(3) Net Load Variability Coefficient (A33)

Net load equals total load minus non-dispatchable renewable power (i.e., wind and solar). The variability coefficient (VC) characterizes grid load variability. As a standardized measure of data dispersion, VC eliminates the influence of magnitude differences.

$$N_i = W_{i,N} + W_{i,T} - W_{i,w} - W_{i,s} \quad (26)$$

$$A_{33} = VC = \frac{\sqrt{1 \Big/ 8760 \times \sum_{i=1}^{8760} \left(N_i - \overline{N}\right)^2}}{\overline{N}} \quad (27)$$

where N_i represents the system's net load at time i; $W_{i,N}$ denotes the system demand at time i; $W_{i,T}$ indicates the external power import; $W_{i,w}$ accounts for wind power generation; $W_{i,s}$ corresponds to photovoltaic generation; and $\overline{N}$ signifies the average net load across all time intervals.

Social indicators.

In comprehensive power system transition assessments, social indicators typically evaluate the system's support and promotion of societal production and living standards across reliability, safety, and adaptability dimensions. This study selects two key indicators: employment opportunities and regional development, quantified through expert survey methodology as detailed below:

(1) Employment Opportunities (A41)

This indicator quantifies the comprehensive employment impacts of different planning schemes, including direct job creation during construction and operation phases, indirect employment effects through supply chain linkages, skill development opportunities, and long-term employment stability. Expert evaluation considers multiple dimensions: construction phase employment intensity, operational workforce requirements, technical skill development potential, and sustainable employment prospects.

(2) Regional Development (A42)

This indicator assesses the broader socio-economic development contributions of power system planning schemes to regional growth, including infrastructure improvement, industrial structure optimization, technological innovation spillovers, and regional competitiveness enhancement. Expert assessment encompasses infrastructure development catalytic effects, industrial clustering potential, innovation ecosystem development, and regional economic diversification impacts.

3. Case study

3.1. Regional overview and parameter settings

3.1.1. Research area overview

The uneven geographical distribution of water, coal, wind, and solar energy resources has led to significant differential characteristics in regions with different water resource endowments in China. The three provinces were strategically selected based on their distinct resource endowment characteristics, which represent the primary pathways of China's low-carbon transition. Xinjiang Uygur Autonomous Region represents the renewable-dominated transition pathway, with abundant solar and wind energy resources providing high renewable energy penetration potential. Despite limited precipitation (150–200 mm/year), Xinjiang possesses significant hydropower potential (28.5 GW) in the Ili River and Altai Mountain regions. The National Pumped Storage Medium- and Long-term Development Plan (2021–2035) identifies 30 potential sites with 39 GW capacity, primarily utilizing existing reservoirs and seasonal snowmelt. The arid climate necessitates closed-loop PSH systems with minimal water consumption (annual evaporation $<$2% of storage volume). Xinjiang demonstrates the transition model for regions shifting from fossil fuel dependence to renewable energy dominance. Shaanxi Province represents the coal-to-clean transition pathway, where coal resources dominate the current power structure. With moderate water resources (43.2 billion m^3/year) concentrated in the northern Yellow River basin and southern Qinling Mountains, the province plans 22 PSH sites (28 GW total) utilizing abandoned mining subsidence areas and mountainous terrain with adequate elevation differences ($>$300 m). Water availability during the dry season (November–March) constrains continuous PSH operation, addressed through seasonal storage scheduling. This province exemplifies the transition challenges faced by coal-dependent regions transitioning toward clean energy systems while maintaining energy security. Yunnan Province represents the flexibility enhancement pathway, relying on its unique water resource endowment and substantial hydropower installations. The province is water-abundant (222.8 billion m^3/year) with extensive hydropower infrastructure (78.3 GW installed, 75% of provincial capacity). The planned 12 GW PSH leverages existing cascade reservoirs on the Lancang, Jinsha, and Honghe rivers, where hybrid PSH-conventional hydropower schemes can share water resources. The region's abundant precipitation (1100–1600 mm/year) ensures year-round water availability with minimal supply constraints. This region demonstrates the pathway for hydropower-dominated areas to enhance system flexibility and support renewable integration.Together, these three provinces cover the main energy transition patterns outlined in China's 14th Five-Year Plan and provide representative insights for other provinces with similar resource characteristics. The resource distribution pattern of the three provinces is shown in Fig. 3.

Hydropower and pumped storage have relatively high flexibility and dispatchability in power generation systems. Traditional hydropower units are typically limited by minimum operating output of approximately 20–30%, but through full-range hydropower flexibility retrofitting, the minimum operating output can be reduced to 0, significantly expanding the regulation range. Meanwhile, variable-speed pumped storage hydropower technology offers significant advantages over traditional fixed-speed pumped storage. Traditional fixed-speed units can only operate at rated speed, whereas variable-speed technology employs AC-DC-AC frequency conversion systems, allowing units to operate within a wide speed range in both pumping and generating modes, demonstrating stronger adaptability in power grid frequency regulation, reserve capacity, and energy storage dispatching.

Based on the above analysis, this paper further explores enhancing system flexibility through hydropower flexibility retrofitting and variable-speed pumped storage units. To more clearly demonstrate the flexibility contribution of hydropower and pumped storage, the related schemes set the hydropower ramping rate at 45% of rated capacity per minute, which is intended to represent a high-end technical capability that could be achieved after flexibility retrofitting and to examine the upper-bound contribution of hydropower as a balancing resource in high-renewable systems, as shown in Table 1. This setting is consistent with international efforts to improve hydropower flexibility; for example, the EU Horizon 2020 HydroFlex project focuses on enabling highly flexible hydropower operation with large ramping rates and frequent start–stop cycling [35]. Related studies also indicate that under more frequent flow-ramping scenarios, corresponding operational strategies and impact-mitigation considerations are necessary, which provides practical support for using “high-ramping/high-cycling” settings in scenario exploration [36]. Meanwhile, to avoid coal capacity providing sufficient flexibility in the scenarios and thereby masking the marginal effects of hydropower and pumped storage, we also reduce a certain amount of coal power installation in the corresponding schemes. The specific schemes are shown in Table 3.

3.1.2. Relevant data preparation

(1)Load demand data.

The load data for Xinjiang, Shaanxi, and Yunnan provinces were obtained from the National Electric Power Dispatching and Communication Center's historical operational database. Based on the provisions outlined in the “Notice on Doing a Good Job in Signing Medium and Long-term Electricity Contracts in 2020” issued by the National Development and Reform Commission, this study extracted and analyzed typical load profiles for working days and holidays in each province. These representative daily load curves were then combined according to

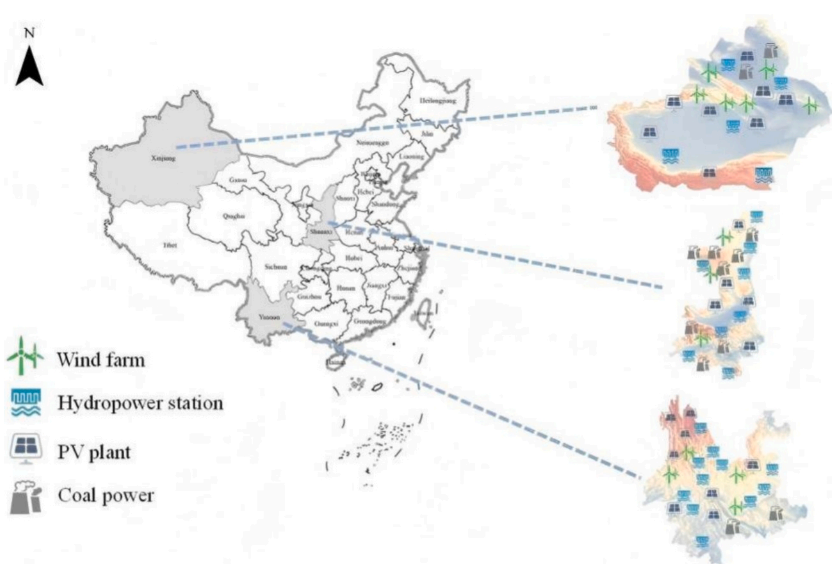

Fig. 3. Energy distribution in provinces with different resource endowments.

Table 3
Flexibility parameters for units of different power generation technologies.

Flexibility Power Generation Technology Type		Stable Output Lower Limit	Ramping Rate (%/min)	Start-up Time (h)	Shutdown Time (h)	Unit Non-rated Efficiency
Coal Power Units	Before retrofitting	0.5–0.6	1–2	6–8	4(6)	0.4
	After retrofitting	0.3–0.35	3–6	4–5	3.5(4)	0.5
Combined Heat and Power Units	Before retrofitting	0.8	1–2	6–9	2(1)	0.25–0.35
	After retrofitting	0.5	3–6	4–5	2(1)	0.3–0.45
Gas Units	Single cycle	0	15	<0.1	0.1(0.2)	0.2–0.4
	Combined cycle	0.2	8	1.1–1.5	0.1(0.2)	0.3–0.45
Hydropower Units		0.4–0.5	60	<0.1	0(0)	0.6–0.8
Pumped Storage Units		±1	67	<0.1	0(0)	0.6–0.8

the annual calendar distribution of working days and holidays to generate the complete 8760-h annual load demand time series for each province. This approach ensures that the seasonal variations, weekly patterns, and holiday effects on electricity consumption are accurately captured in the simulation.

(2) Cost parameters and flexibility parameters.

The cost structure of the power generation system encompasses both investment costs and operational costs. Investment costs include one-time capital expenditures for power generation technologies, while operational costs comprise start-up cost, maintenance cost, variable cost, and peak-shaving cost. The loss-of-load cost is determined by the production-to-electricity ratio method, calculating the unit cost (*10^4 CNY/MW*) through the ratio of regional gross domestic product to total social electricity consumption.

For investment cost calculation, we establish a comprehensive cost parameter framework covering all power generation technology types with explicit data sources and calculation methods. For pumped storage hydropower (PSH), we use fixed-speed PSH as the baseline (8000 *CNY/kW*) based on China's national planning documents and recent engineering projects. Variable-speed PSH technologies incur additional investment costs due to advanced power electronics, frequency conversion systems, and enhanced control capabilities. Following the rigorous cost decomposition methodology of Ref. [9], the cost premiums are calculated as follows: Variable-speed doubly-fed induction generator (DFIG) PSH: Additional cost +7% over fixed-speed (Total: 8560 *CNY/kW*). For hydropower flexibility retrofitting, we estimate a unit cost of 1500 *CNY/kW* based on engineering consultation and industry practice. This cost accounts for upgraded turbine governors with faster response times, expanded operating range modifications, enhanced guide vane control systems, and improved start-up/shutdown procedures.For model tractability and focus on technology comparison, we adopt unified cost parameters across the three provinces rather than region-specific values. While actual construction costs vary by location due to geological conditions, labor costs, and material transportation distances (typically ±10–15% variation), using standardized costs allows for clearer comparison of technology alternatives and planning strategies across regions with different resource endowments.This simplification is justified for our research objective of comparing hydropower-pumped storage configurations rather than conducting detailed project-level feasibility analysis. Regional cost variations are acknowledged as a limitation in Section 5 (Discussion), and we recommend that actual project planning incorporate site-specific cost assessments.

Unit flexibility parameters reflect the start-stop and rapid regulation capability of the unit, mainly including stable output lower limit, ramping rate, shutdown time, start-up time, and unit non-rated efficiency, etc. These parameters can be optimized through flexibility enhancement retrofitting, thereby improving the system's ability to adapt to load demand changes. Tables 2a, 2b and 3 provide the relevant parameter values of the power generation system.

3.2. *Study on flexibility enhancement effect under energy transition measures*

3.2.1. *Capacity expansion planning results*

Based on the initial data of various energy installed capacities in the three provinces in 2021, through the calculation results of the power generation system planning model, this study obtains the system installation data of Xinjiang Uygur Autonomous Region, Shaanxi Province, and Yunnan Province in 2030 (transition period) and 2060 (deep decarbonization period), providing a data foundation for exploring hydropower and pumped storage planning schemes under the background of power system flexibility enhancement. Fig. 4 shows in detail the distribution of installed capacity of various power generation technologies in the three provinces, and the related calculation method is detailed in Appendix B.

From Fig. 4, it can be seen that the 2021 base year data shows that coal power installations in Xinjiang and Shaanxi reached 45.2GW and 38.7GW respectively, significantly higher than other energy sources; Yunnan is dominated by hydropower, with an installation of 75.6GW. Xinjiang has significant advantages in wind and solar resources, with installations of 22.6GW and 13.8GW respectively.

By the transition period of 2030, although the total installed capacity of coal power in the three provinces has increased to 184.7GW (97.5 + 70.5 + 16.7), its proportion has significantly decreased from the total system capacity. To support the energy transition and provide system flexibility, Xinjiang, Shaanxi, and Yunnan have planned 14.4GW, 13.8GW, and 11.6GW of pumped storage installations respectively. During the same period, wind and solar installations have increased significantly to 117.8GW and 99.3GW respectively across the three provinces, mainly benefiting from the improved economics of renewable energy and the promotion of carbon reduction policies.

In the deep decarbonization stage of 2060, the proportion of coal power in the three provinces has dropped to extremely low levels, with Yunnan having only 3.6GW left, while Xinjiang retains 21GW and Shaanxi 20.7GW. Pumped storage planning has been fully implemented, with installations in Xinjiang, Shaanxi, and Yunnan reaching 37GW, 34.4GW, and 25.9GW respectively, mainly due to their demand for system flexibility after large-scale intermittent renewable energy grid connection (solar power reaching 1622.8GW total), as well as suitable topographical conditions and the important role of pumped storage in peak regulation and frequency modulation.

3.2.2. *Hydropower and pumped storage flexibility analysis*

Based on the scenarios set in Section 3.2.1, this section conducts optimization simulation of power generation system operation in 2030 and 2060 under coal power retirement scenarios in the three provinces. The power system operation results are shown in Fig. 5 and Fig. 6.

In 2030, all three provinces demonstrate effective supply-demand balance with renewable energy providing substantial contributions during daytime hours. Xinjiang shows the highest renewable penetration with wind and solar generation dominating the energy mix, while

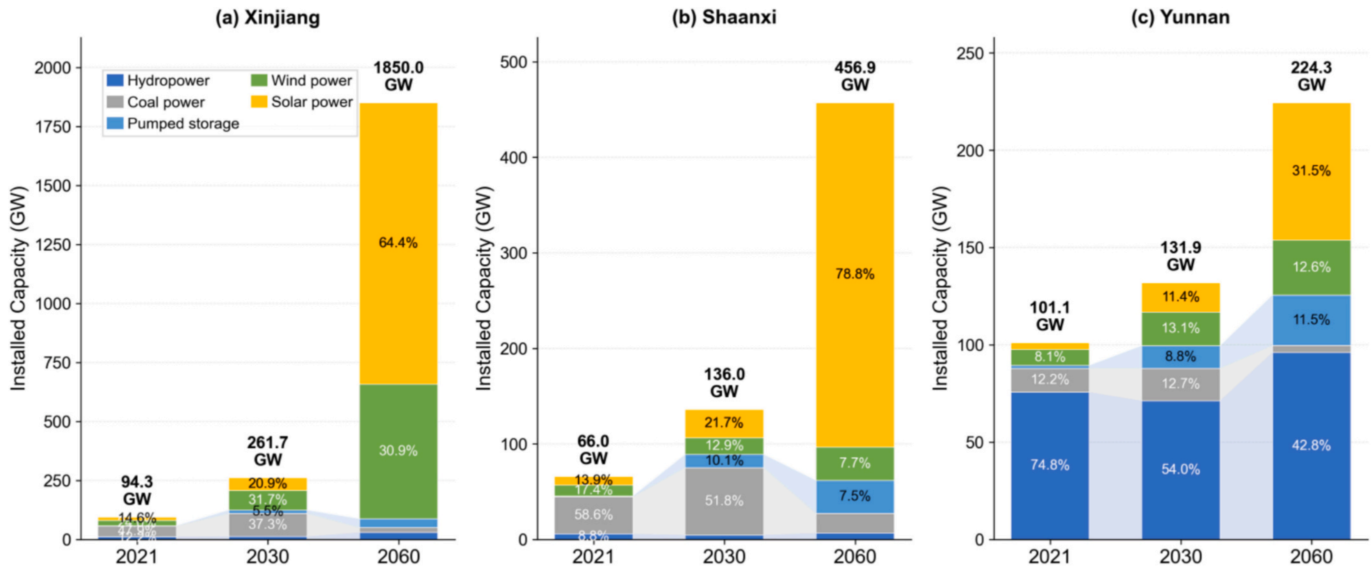

Fig. 4. Capacity expansion planning results for each province.

Shaanxi maintains a more balanced portfolio with coal power providing baseload support, and Yunnan leverages its abundant hydropower resources as the primary generation source. Pumped storage pumping (negative areas) occurs strategically during periods of excess renewable generation, particularly during midday solar peaks. By 2060, the energy landscape transforms dramatically with renewable sources becoming the dominant generation technologies across all provinces. Coal power is substantially reduced in all scenarios, while pumped storage operations become more frequent and intensive to manage increased renewable variability. The analysis reveals distinct regional characteristics: Xinjiang achieves near-complete renewable energy dominance through wind-solar complementarity, Shaanxi experiences the most significant transformation from coal-dependent to renewable-dominated system, and Yunnan optimizes the synergy between its massive hydropower base and expanding solar capacity. The coordinated operation of pumped storage and flexible hydropower successfully manages renewable intermittency without curtailment, demonstrating the effectiveness of the proposed integrated planning approach.

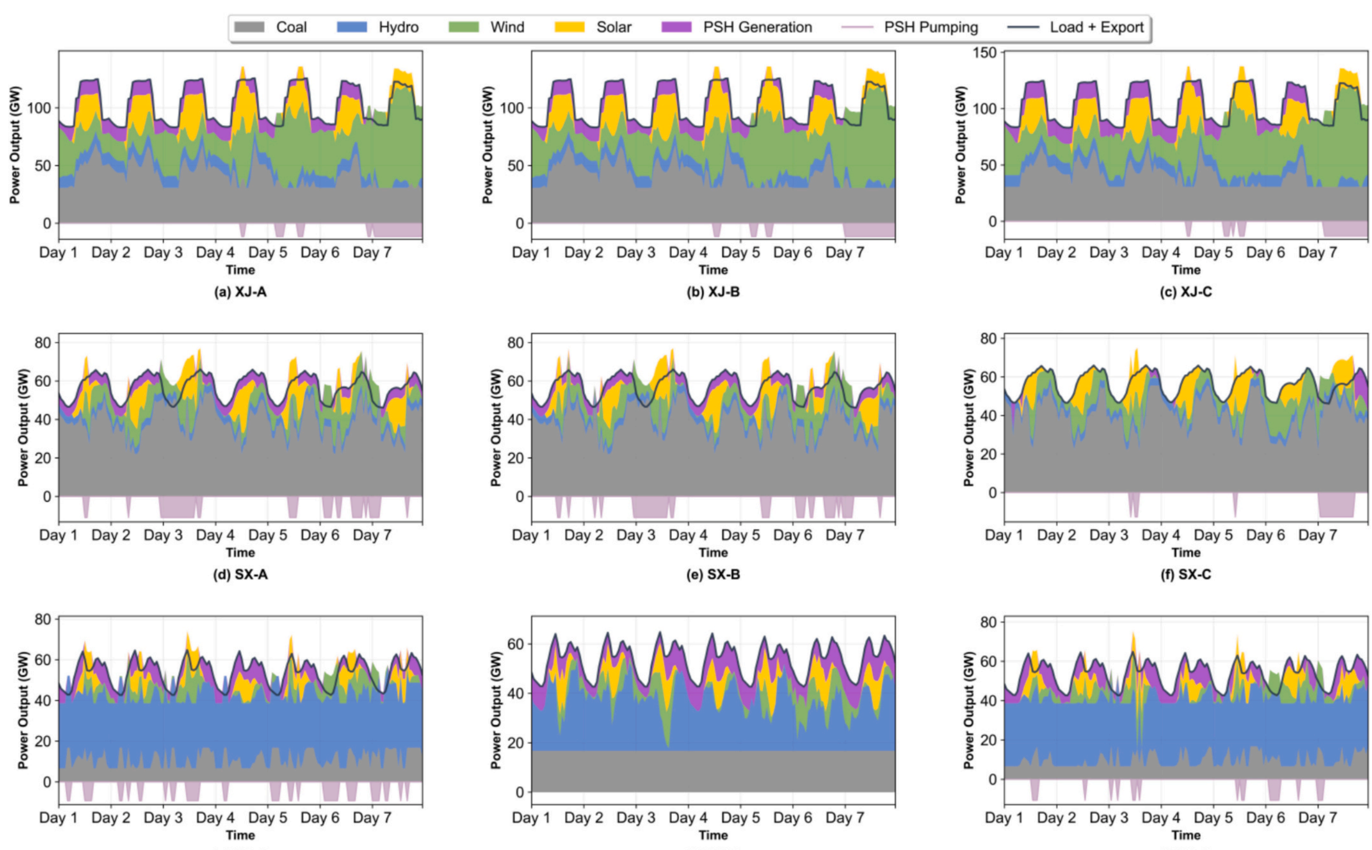

Fig. 5. Typical weekly output of three different scenarios in three provinces in 2030.

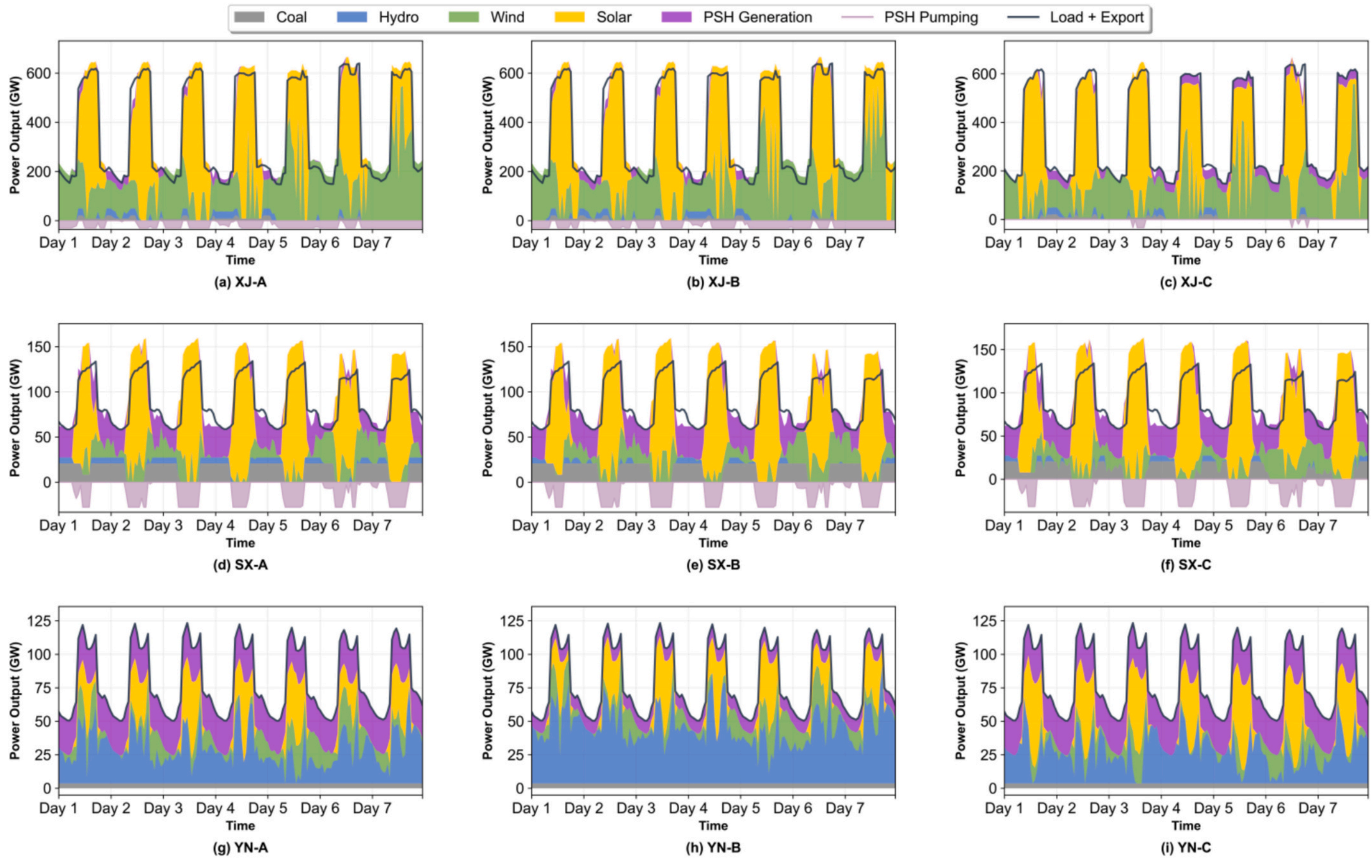

Fig. 6. Typical weekly output of three different scenarios in three provinces in 2060.

The specific annual loss-of-load hours and annual cumulative loss-of-load for each scenario are shown in Table 4.

The 2030 data shows that through hydropower flexibility retrofitting, the loss-of-load hours in Xinjiang, Shaanxi, and Yunnan decreased by 6, 20, and 0 times respectively, and the annual loss-of-load amount decreased by 3686 MWh, 29159 MWh, and 0 MWh. In the fixed-to-variable speed pumped storage comparison scenario, the stronger flexibility of variable-speed units reduced the loss-of-load hours in the three provinces by 40, 166, and 0 times, and the annual cumulative loss-of-load decreased by 326042 MWh, 1182759 MWh, and 1.21 MWh. For Yunnan, under the scenarios of hydropower flexibility retrofitting and variable-speed pumped storage, the loss-of-load hours in 2060 decreased by 9 and 90 times respectively, and the annual cumulative loss-of-load decreased by 1823.8 and 651,557.4 MWh, showing that variable-speed pumped storage has a more significant enhancement effect. For Xinjiang and Shaanxi, due to the small proportion of hydropower and more full-load operation time, the effect of hydropower flexibility retrofitting is limited, but in the variable-speed pumped storage scenario, the loss-of-load hours decreased by 131 and 142 times respectively, and the annual cumulative loss-of-load significantly decreased. The 2060 data further verifies this trend, indicating that as the proportion of clean energy increases, the role of variable-speed pumped storage in enhancing system flexibility will become more prominent.

Table 4
The number of loss-of-load hours and the annual loss-of-load cumulative amount under three scenarios in three provinces in 2030 and 2060.

Scenario	Loss-of-load hours	Annual cumulative loss-of-load (MWh)	Scenario	Loss-of-load hours	Annual cumulative loss-of-load (MWh)
XJ-A	172	971,812	XJ-A-2	993	105,627,236
XJ-B	166	968,126	XJ-B-2	993	105,626,936
XJ-C	132	647,570	XJ-C-2	862	52,539,037
SX-A	657	4,014,600	SX-A-2	3382	61,640,895
SX-B	637	3,985,441	SX-B-2	3382	61,637,595
SX-C	491	2,831,841	SX-C-2	3240	60,573,260
YN-A	2	104.54	YN-A-2	126	776,884.4
YN-B	2	104.54	YN-B-2	117	775,060.6
YN-C	2	102.36	YN-C-2	36	125,327.5

To analyze the system's utilization effect of wind and solar renewable energy, the wind and solar utilization rate indicator is cited to explain it, and the wind and solar utilization rate change graph is drawn as shown in Fig. 7.

From Fig. 7, it can be seen that the wind and solar utilization rates in Xinjiang and Shaanxi provinces under hydropower flexibility retrofitting and variable-speed pumped storage scenarios show an increasing trend, with Xinjiang increasing by 0.6% and 5% in the two scenarios respectively, and Shaanxi increasing by 1.1% and 1.2% in various scenarios. Among them, Yunnan Province has a high level of wind and solar utilization, with a wind and solar utilization rate of 99.9%. The above changes indicate that in 2030, through enhancing hydropower flexibility or pumped storage hydropower technology, it is conducive to promoting the utilization and consumption of wind and solar resources. From Fig. 7, it can be seen that the wind and solar utilization rates in Xinjiang, Shaanxi, and Yunnan provinces under hydropower flexibility retrofitting and variable-speed pumped storage scenarios show an increasing trend, with increases of 0.6%, 1%, and 2% respectively under the hydropower flexibility retrofitting enhancement scenario, and increases of 1.2%, 3.2%, and 17.8% respectively under the variable-speed pumped storage scenario. The overall trend changes are consistent with 2030, but the enhancement effect in Yunnan Province in 2060 is relatively more obvious, which is related to the increase in its wind and solar installations.

In summary, by 2060, despite the hydropower unit flexibility

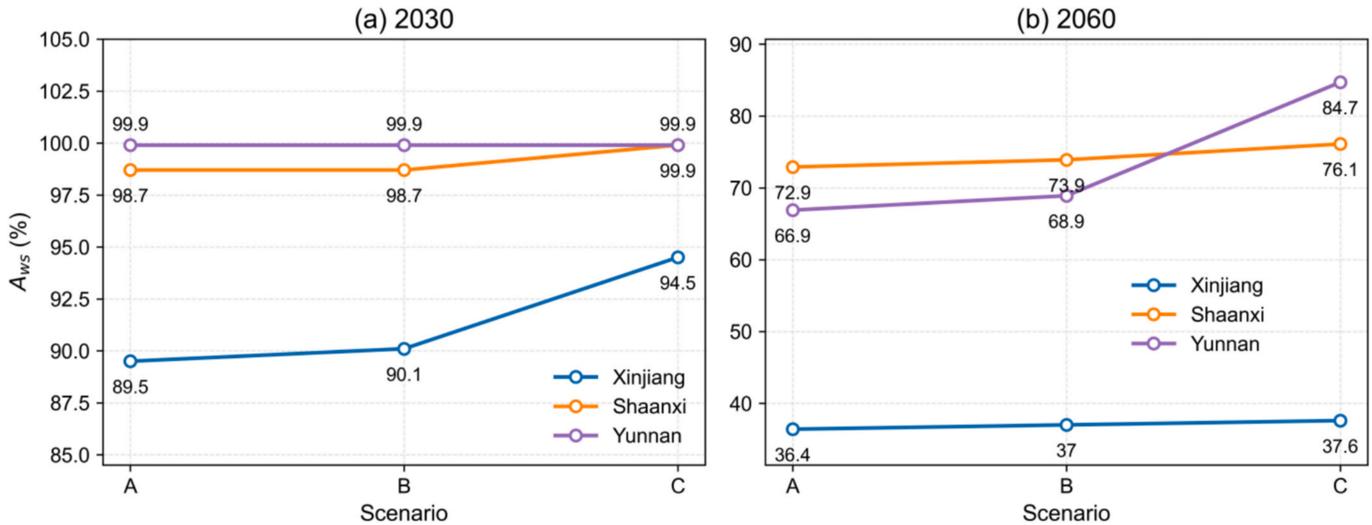

Fig. 7. The utilization rate of different scenes in three provinces in 2030.

retrofitting and variable-speed pumped storage reconstruction in Xinjiang and Shaanxi, they have still not been able to effectively solve the problem of mismatch between system output and demand. This is mainly because the planning model only considers resource development, carbon emission limits, and clean energy proportion targets, leading to a too rapid reduction in adjustable coal power, while hydropower and pumped storage, limited by resources, have not been able to completely replace the flexibility potential of coal power, reducing system flexibility. Given the limited effects of existing hydropower flexibility retrofitting and pumped storage variable-speed technology enhancement, the next step will explore schemes to meet demand by increasing pumped storage capacity.

3.2.3. Pumped storage incremental planning analysis

Regarding the limited effect of system hydropower flexibility enhancement in the above schemes, the system output characteristics of Xinjiang and Shaanxi in 2060 are mainly manifested as photovoltaic not generating electricity at night, causing regulatory power sources to be unable to fill the system demand gap. This section will explore reducing system loss-of-load through pumped storage incremental planning in the power generation system, and specifically using power loss probability and loss-of-load probability indicators to determine the minimum pumped storage planning capacity required by the system.

Based on the existing power installation foundation in Xinjiang and Shaanxi, this paper explores the minimum installation demand to achieve power loss probability and loss-of-load probability below 5% by gradually increasing pumped storage capacity (3GW each time), to meet system reliability requirements and provide scientific basis for policy making. Table 5 shows various planning schemes that meet the indicator restrictions.

From Table 5, it can be seen that as pumped storage installation gradually increases to 60GW, the system's power loss probability and loss-of-load probability indicators decrease from 9.84% to 5.2% and from 5.69% to 4.12% respectively. When the pumped storage installation reaches 63GW, the system's power loss probability and loss-of-load probability are 4.82% and 3.97% respectively. For Shaanxi, as pumped storage installation gradually increases to 69GW, the system's power loss probability and loss-of-load probability indicators decrease from 36.99% to 5.14% and from 6.99% to 0.958% respectively. When the pumped storage installation reaches 72GW, the system's power loss probability and loss-of-load probability are 4.23% and 0.817% respectively. The above indicates that when the pumped storage installation in the system increases, it can effectively reduce the system's loss-of-load hours and total loss-of-load, improving the system's flexibility and stability.

Table 5
Xinjiang, Shaanxi pumped storage capacity increment planning related indicators in 2060.

Scenario	Pumped Storage Installation/GW	Power Loss Probability/%	Load-of-Loss Probability/%
XJ-D	60	5.20	4.12
XJ-E	63	4.82	3.97
SX-D	69	5.14	0.958
SX-E	72	4.23	0.817

In summary, to meet the system's power loss probability and loss-of-load probability indicator conditions, the system pumped storage planning installation in Xinjiang and Shaanxi provinces needs to reach 63GW and 72GW respectively. In areas with a small proportion of hydropower, when hydropower flexibility retrofitting enhancement cannot meet the system flexibility insufficiency problem caused by the reduction of coal power resources. Compared with the Medium and Long-term Development Plan for Pumped Storage (2021–2035), the 2060 optimized capacities for Xinjiang (63 GW) and Shaanxi (72 GW) exceed the national 2035 targets (39 GW and 56 GW) by 62% and 29%, respectively, while Yunnan's result (15.2 GW) is 16% below the national target (18 GW). These discrepancies primarily reflect different time horizons (2035 mid-term vs. 2060 carbon neutrality) and optimization objectives. Accordingly, the 2060 results should be interpreted as technical upper-bound demands under deep decarbonization scenarios rather than implementable planning targets, complementing the near-term targets in national planning.

3.2.4. Multi-dimensional low-carbon transition benefit comprehensive evaluation

To evaluate the characteristics and rationality of various schemes in economic, environmental, technical, and social aspects, this section first briefly evaluates the differences between various schemes and analyzes the causes of the differences. Finally, it comprehensively evaluates the results of various schemes, comparing the comprehensive scores between various schemes, in order to provide reference for relevant hydropower and pumped storage planning schemes.

This section analyzes the economic feasibility and benefits of the schemes through two indicators: construction investment cost and operation cost. At the same time, it selects wind and solar utilization rate and CO_2 emissions to evaluate wind and solar resource consumption and carbon emissions; adopts power loss rate, loss-of-load probability, and net load variability coefficient to analyze the sustainability and safety stability of the power generation system. In addition, through questionnaire surveys, it compares the employment opportunities and regional development situations under five schemes. Detailed results are shown in Fig. 8.

From an economic perspective, construction investment costs and operation costs vary by region. Taking 2030 as an example, the investment costs of hydropower flexibility retrofitting scenarios in Xinjiang and Yunnan are 1590 billion yuan and 1230 billion yuan respectively, higher than the variable-speed pumped storage scenario; while Shaanxi is the opposite, with the variable-speed pumped storage scenario costing more in 2060. In terms of operation costs, in 2030, Xinjiang and Shaanxi have the lowest costs in the hydropower flexibility retrofitting scenario, while Yunnan has the lowest in the basic scenario; by 2060, the variable-speed pumped storage operation costs are generally the lowest. From the technical indicators, the variable-speed pumped storage scenario has the lowest system carbon emissions, such as 409.138Mt in Xinjiang in 2030, decreasing to 42.874Mt in 2060. The wind and solar utilization rate perform better in the variable-speed pumped storage scenario, with Xinjiang increasing by 5% in 2030, and as pumped storage planning increases, it can increase to about 60%. Power loss rate and loss-of-load probability decrease as system flexibility enhances, with hydropower flexibility retrofitting reducing Xinjiang's power loss rate from 1.96% to 1.89% in 2030. In 2060, as pumped storage installation increases, Xinjiang's power loss rate can decrease to 5.2054%, indicating that increasing adjustable power sources improves power supply reliability. In terms of net load variability coefficient, after increasing pumped storage capacity, Xinjiang and Shaanxi decreased from 83.7506% to 75.9116% and from 64.5338% to 61.3921% respectively, indicating that pumped storage helps smooth out net load fluctuations. For employment opportunities, whether in 2030 or 2060, the three provinces of Xinjiang, Shaanxi, and Yunnan have higher employment opportunities and regional development in the retrofitting scheme with hydropower ramping at 45% and operating lower limit at 0 than in other schemes, at 6.67 and 7 respectively. The incremental pumped storage scheme provides the best employment opportunities and regional development for Xinjiang and Shaanxi, at 6.67 and 8. Overall, the variable-speed pumped storage scheme performs well in economic benefits, environmental effects, system stability, and social benefits.

3.3. Comprehensive evaluation

To reflect the advantages and disadvantages of hydropower and pumped storage planning schemes under the low-carbon transition of the power generation system, the CRITIC-TOPSIS method is used to calculate the weights of various schemes. Subsequently, trough the percentage scoring method, the comprehensive scores of various planning schemes are drawn.

First, the weight proportions of indicators in various situations in the three provinces are analyzed, calculating the weights of the results of hydropower and pumped storage planning schemes in the three provinces in 2030 and 2060, uniformly classified as shown in Table 6.

As shown in Table 6, there are differences in the weights of economic, technical, environmental, and social indicators for the planning schemes of the three provinces in 2030 and 2060. Among them, the weights of economic and technical indicators are relatively high, for example, in the comparison of scenarios in the three provinces in 2030, the economic indicator weight proportion in the Shaanxi scenario is the highest, reaching 0.296, and between the two, the construction investment cost weight is slightly higher; in the Yunnan scenario scheme, the technical indicator reaches 0.361, and the economic indicator weight is the lowest, reaching 0.2078. Looking at the 2060 indicators, in the economic indicators, the construction investment cost weight in Shaanxi and Xinjiang is higher than the system operation cost, related to their higher construction investment installation; while Yunnan Province has the lowest economic indicator weight, at 0.2139.

To analyze the comprehensive scores of various schemes and their score proportions in various indicators, the 2030 and 2060 comprehensive evaluation scheme percentage relative score chart is drawn, as shown in Fig. 9.

Analysis of the comprehensive evaluation results of various schemes in the three provinces in 2030 and 2060 shows significant differences. In 2030, the variable-speed pumped storage scenarios in Xinjiang and Shaanxi (XJ-C, SX-C) scored the highest, at 99 and 99 respectively, while Yunnan Province's basic scenario (YN-A) scored the highest at 99. Looking at various indicators, Xinjiang's environmental indicators are generally low (11.44–18.3), while Shaanxi's economic indicators perform well (24.45–29.7). By 2060, the evaluation results have changed significantly: Xinjiang and Shaanxi's incremental pumped storage planning scenarios (XJ-E, SX-E) perform best, scoring 99 and 93.7 respectively, while Yunnan's variable-speed pumped storage scenario (YN-C-2) scores the highest at 99. In terms of indicator weights, Xinjiang and Shaanxi pay more attention to economic indicators, with the latter even accounting for over 40 of the total score in Shaanxi; while Yunnan emphasizes technical indicators, such as the technical score (30.25) in the YN-C-2 scenario being higher than the other three types of indicators. These differences reflect the different resource endowments and development strategies of each province: Xinjiang and Shaanxi focus more on economic benefits, while Yunnan emphasizes technological improvement. The performance of the same scenario in different provinces also varies, such as the hydropower flexibility retrofitting scenario scoring 83.42, 75.16, and 79.75 in Xinjiang, Shaanxi, and Yunnan respectively, which is closely related to the economic, technical level, and environmental performance of the system operation in each province.

4. Conclusion

This study developed a collaborative modeling framework to systematically quantify the effects of hydropower flexibility retrofitting and pumped storage planning on the low-carbon transition of power systems in Chinese provinces with diverse water resource endowments. By integrating long-term capacity expansion, detailed operational simulation, and a multi-dimensional evaluation system, we systematically compared different flexibility enhancement strategies for Xinjiang, Shaanxi, and Yunnan.

The key findings reveal that both hydropower flexibility retrofitting and the deployment of variable-speed pumped storage significantly enhance system stability and renewable energy integration, though their effectiveness varies based on regional context. Variable-speed pumped storage consistently demonstrates superior performance in reducing loss-of-load events, with reductions of up to 166 h in Shaanxi, far exceeding those from hydropower retrofitting. For regions with limited hydropower resources like Xinjiang and Shaanxi, simply retrofitting existing assets is insufficient to meet the flexibility demands of a deeply decarbonized grid. Our analysis indicates that substantial new pumped storage capacity—63 GW for Xinjiang and 72 GW for Shaanxi by 2060—is required as a technical upper-bound demand to maintain system reliability under deep decarbonization constraints, exceeding the national 2035 planning targets (39 GW and 56 GW) by 62% and 29%, respectively, and complementing near-term implementable targets.

Furthermore, the comprehensive evaluation underscores that optimal planning strategies are highly region-specific. In the 2030 transition period, variable-speed pumped storage schemes are preferable for the renewable- and coal-dominated provinces of Xinjiang and Shaanxi, while the basic scenario remains optimal for hydropower-rich Yunnan. By 2060, however, strategies shift, with incremental pumped storage becoming optimal for Xinjiang and Shaanxi, and a variable-speed scheme becoming the best choice for Yunnan. These findings highlight the necessity of tailoring flexibility investment pathways to local resource endowments and evolving system needs.

In conclusion, this research provides critical, empirically grounded

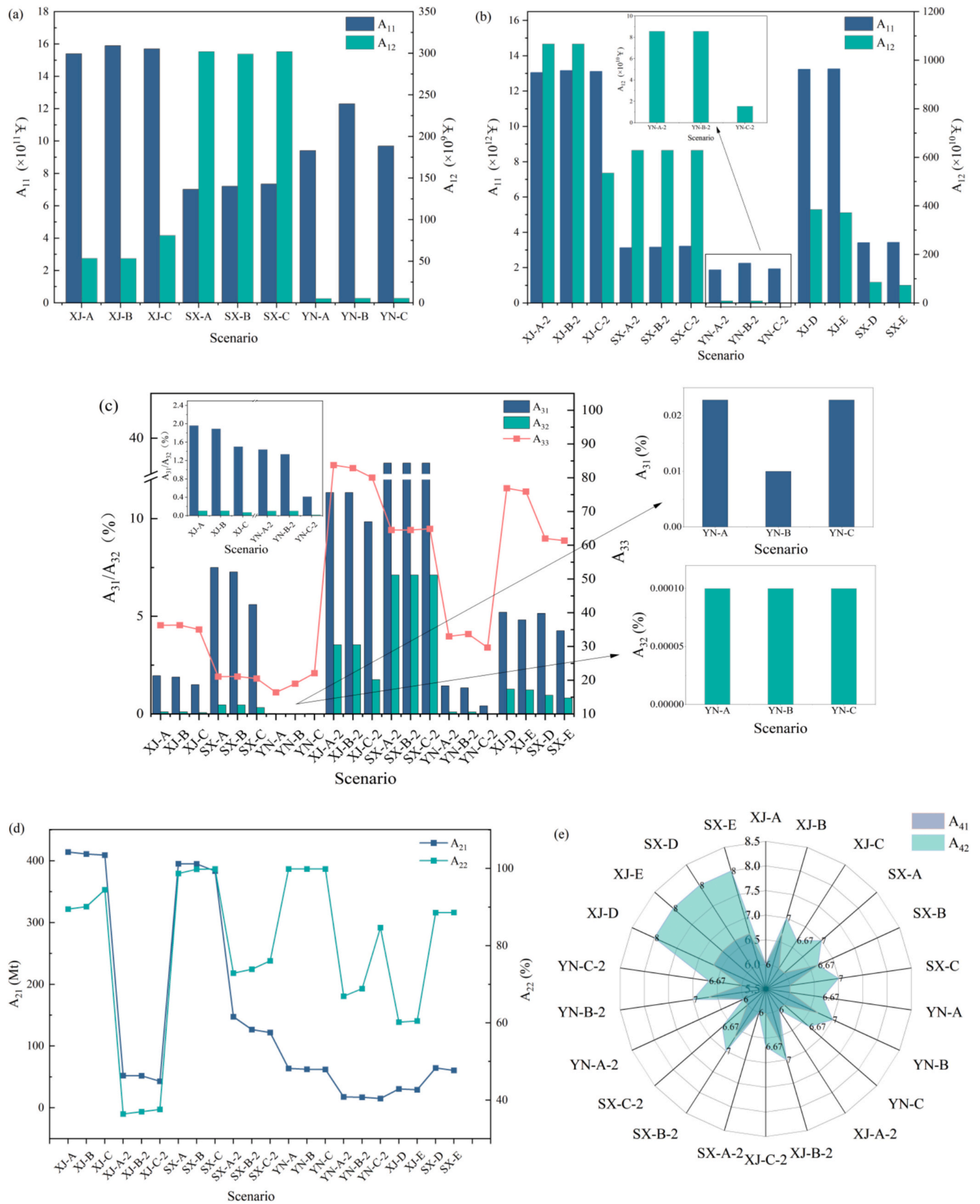

Fig. 8. Economic, environmental, technological, and social indicators of each planning scheme.

Table 6
Evaluation weights of each program index in the three provinces in 2030.

	A_{11}	A_{12}	A_{21}	A_{22}	A_{31}	A_{32}	A_{33}	A_{41}	A_{42}
XJ (30)	0.1371	0.1418	0.0883	0.0947	0.0941	0.0789	0.0896	0.1747	0.1008
SX (30)	0.1516	0.1454	0.0876	0.0939	0.0942	0.0939	0.0692	0.1825	0.0818
YN (30)	0.0883	0.1195	0.1501	0.0882	0.1020	0.1049	0.1096	0.1276	0.1098
XJ (60)	0.3013	0.0708	0.0799	0.0709	0.0994	0.0720	0.0915	0.1201	0.0943
SX (60)	0.376	0.076	0.076	0.076	0.0718	0.0683	0.0724	0.1135	0.007
YN (60)	0.0585	0.1554	0.1133	0.1536	0.0965	0.0959	0.1101	0.1559	0.0608

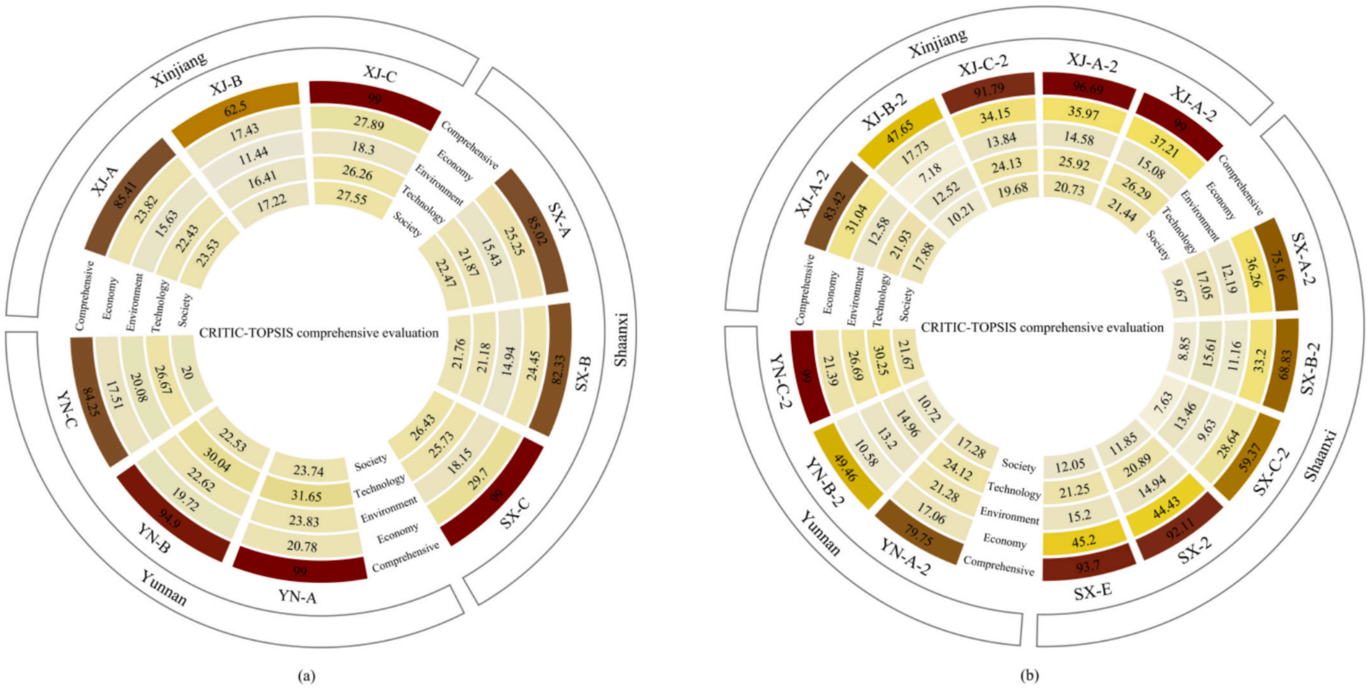

Fig. 9. Comprehensive evaluation results of hydropower and pumped storage planning schemes in three provinces: (a) 2030 transition period; (b) 2060 deep decarbonization period.

insights for policymakers, demonstrating that a one-size-fits-all approach to flexibility planning is inadequate. By quantifying the distinct benefits of different hydropower and pumped storage technologies, this study offers a robust analytical basis for formulating differentiated, resource-adaptive strategies to support China's secure and cost-effective transition to a low-carbon power system.

5. Discussions

Despite the comprehensive approach taken in this study, several limitations should be acknowledged. The deterministic nature of the modeling framework does not fully capture the uncertainties in renewable energy generation, load forecasting, and water resource availability. The models assume perfect foresight in system operation, whereas real-world power systems face significant forecasting challenges. The study focuses primarily on technical and economic aspects of flexibility enhancement, with simplified representations of environmental impacts. While CO_2 emissions are considered, other environmental factors such as water consumption, land use changes, and ecosystem impacts of hydropower and pumped storage are not comprehensively addressed.

Regarding inter-provincial power exchange modeling, this study adopts an exogenous transmission assumption where transmission volumes are treated as pre-determined parameters based on historical data. We do not model endogenous optimization of transmission volumes responding to changing capacity mix. With China's advancing electricity market reform toward a unified national electricity market, future inter-provincial trading will shift from contract-based to market-based bidding, enabling cross-provincial flexibility resource sharing. Ignoring market-based cross-provincial flexibility sharing may overestimate provincial PSH demand by 15–30%. Accordingly, the 2060 optimized capacities (63 GW for Xinjiang, 72 GW for Shaanxi, and 15.2 GW for Yunnan) represent technical upper-bound demands under provincial self-sufficiency assumptions, exceeding the national 2035 planning targets for Xinjiang and Shaanxi by 62% and 29% respectively while Yunnan remains 16% below. These upper bounds serve as long-term baseline references that complement near-term implementable targets for gradual market reform and planning adjustments beyond 2035.

The methodological framework and findings of this study have varying degrees of applicability to other regions and countries. The three-step collaborative modeling approach combining long-term planning with detailed operational simulation is broadly applicable to diverse energy systems globally. The findings regarding the effectiveness of hydropower flexibility retrofitting versus variable-speed pumped storage are most directly applicable to countries with similar geographical and resource characteristics. Countries with significant hydropower resources but different seasonal water availability patterns (e.g., Brazil, Norway, or Canada) may experience different flexibility enhancement effects than observed in China's provinces. It should be noted that the primary methodological contribution of this study lies in application-oriented system integration — specifically addressing previously unexamined technical challenges in cross-scale model consistency, variable-speed PSH parameterization, and hydropower retrofit modeling — rather than paradigm-level methodological innovation. Nevertheless, these implementation-level bottlenecks represent concrete gaps that have constrained quantitative flexibility assessment of advanced pumped storage technologies in prior regional planning literature.

Based on the findings and limitations of this study, several promising directions for future research emerge. Future research should expand beyond ramping rate enhancements to explore comprehensive hydropower flexibility retrofitting options, including wide-load operation capabilities, faster start-stop cycles, and advanced governor control systems. The complementarity between hydropower, pumped storage, and other flexibility resources such as battery storage, hydrogen storage, and demand response should be systematically investigated. Future studies should incorporate climate change scenarios to assess how changing precipitation patterns and water availability might affect the flexibility potential of hydropower resources over the long-term planning horizon to 2060.

CRediT authorship contribution statement

Ziwen Zhao: Writing – review & editing, Writing – original draft, Visualization, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jianling Li:** Data curation. **Xianan Sun:** Conceptualization. **Yong Liu:** Writing – original draft, Investigation, Funding acquisition, Data curation, Conceptualization. **Meng Zhang:** Resources. **Ming Li:** Writing – review & editing, Visualization, Resources. **Xiao Wang:** Writing – review & editing, Resources, Investigation, Data curation. **Kafait Ullah:** Writing – review & editing, Visualization, Resources. **Diyi Chen:** Writing – review & editing. **Hany M. Hasanien:** Writing – review & editing, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apenergy.2026.127750.

Data availability

Data will be made available on request.

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